

Complex Analysis PhD Exam, May 1992

Do all problems. Show all the necessary details so partial credit can be properly assessed.

1. Construct a conformal map which will take the intersection of the disks $|z| < 1$ and $|z - 1| < 1$ to the unit disk.

2. Evaluate $\int_0^\infty \frac{\log x}{1+x^4} dx$ using contour integration.

3. **Warning: Suspected error.** The source reading is clear, but the hypotheses and conclusion are mathematically inconsistent: $\sin \pi z$ has derivative π , not 1, at 0 and is unbounded on the imaginary axis. The intended correction is not uniquely determined, so the source text has been retained.

Let $f(z)$ be entire of order 1 and the zero set of f , $Z(f) = \{0, \pm 1, \pm 2, \pm 3, \dots\}$. Suppose $f'(0) = 1$, $f(\frac{1}{2}) = 1$, and f is bounded on the imaginary axis. Show that $f(z) = \sin \pi z$.

4. Let $P_n(z) = \frac{1}{n!} \frac{d^n}{dz^n} [z^n(1-z)^n]$ be the n^{th} -Legendre polynomial (of degree n). Show that $P_n(-1) \leq (1 + \sqrt{2})^{2n}$. (Hint: Write $P_n(-1) = \frac{1}{2\pi i} \int_C \frac{\zeta^n(1-\zeta)^n}{(\zeta+1)^{n+1}} d\zeta$ over a suitable circle centered at -1 .)

5. This problem has two parts.

- (a) Let f be entire and $f(z) \rightarrow \infty$ as $z \rightarrow \infty$. Show that f is a polynomial.
- (b) If in part (a) we replace f entire by f analytic in $\mathbb{C} - \{0\}$, then what can be said?

6. Let $f(z)$ be analytic in a domain D and on its boundary C . If $|f(z)|$ is constant on C , show that unless $f(z)$ reduces to a constant there must be at least one zero of $f(z)$ in D .

7. Show that the sum of the series

$$\frac{1}{1-z} - \frac{z}{1-z^2} - \frac{z^2}{1-z^4} - \frac{z^4}{1-z^8} - \frac{z^8}{1-z^{16}} - \dots$$

is 1 when $|z| < 1$, but is 0 when $|z| > 1$. This seems to contradict analytic continuation. How is this possible?

8. If $f(z)$ is analytic for $|z| < 1$ and $|f(z)| < \frac{1}{1-|z|}$ ($|z| < 1$), show that the coefficients a_n in the expansion

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

are subject to the inequality

$$|a_n| \leq (n+1) \left(1 + \frac{1}{n}\right)^n < e(n+1).$$

9. This problem has two parts.

- (a) Show that the function

$$w = \left(\frac{1+z}{1-z} \right)^{\frac{2\alpha}{\pi}}, \quad 0 < \alpha < \pi,$$

maps $|z| < 1$ onto the angular sector $|\arg\{w\}| < \alpha$.

- (b) Prove that if $f(z)$ is analytic in $|z| < 1$ and satisfies $f(0) = 1$ and $|\arg\{f(z)\}| < \alpha$, then $|f'(0)| < \frac{4\alpha}{\pi}$.